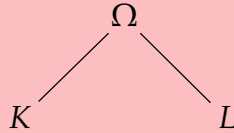


Problem 1:

Let Ω be a field and let K and L be subfields of Ω .



Let KL denote the smallest subfield of Ω containing both K and L . KL is called the *compositum* of K and L . Give an explicit description of the elements in KL .

Let $S = \{\sum_{i=1}^n k_i l_i \mid n \in \mathbb{N}, k_i \in K, l_i \in L\}$

Let $s, t \in S$ then

$$s = \sum_{i=1}^n a_i b_i \quad \text{and} \quad t = \sum_{j=1}^m c_j d_j$$

Thus,

$$st = \sum \sum (a_i c_j)(b_i d_j) \in S$$

Also, $st = 0$ only if s or t is zero since K, L are subfields of Ω . Thus, S is also multiplicative.

Thus, S is an integral domain. Thus, consider the fraction field of S . Call this F .

First, $K \subset F$ and $L \subset F$ since $1 \cdot k \in F$ and $1 \cdot l \in F$ for all $k \in K$ and $l \in L$.

Thus, $KL \subset F$.

Let E be a subfield of Ω s.t. $K \subset E$ and $L \subset E$.

Since E is a field it is closed under addition and multiplication. Thus, for any $a_i \in K$ and $b_i \in L$, each product $a_i b_i \in E$. Thus, we get that any finite sum of such products is also in E . Thus, $S \subset E$.

Since E is a field, E is also closed under inverses except for 0. Thus, if $t \in S$ and $t \neq 0$ then $t^{-1} \in E$.

Thus, $st^{-1} \in E$ for all $s, t \in S$ and $t \neq 0$.

Therefore, $F \subset E$.

Thus, $F = KL$.

Problem 2:

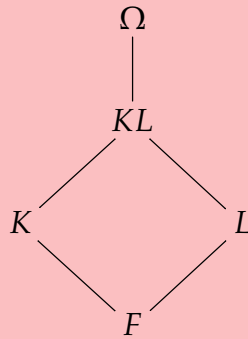
A class of extensions is said to be distinguished if it satisfies the following three properties:

- i. Given a tower of extensions $L/K/F$:

$$\begin{array}{c} L \\ | \\ K \\ | \\ F \end{array}$$

the extension L/F is in the class if and only if L/K and K/F are in the class.

- ii. Given a diagram of field extensions as below,



if K/F is in the class, then KL/L is in the class.

- iii. If K/F and L/F are in the class, and K and L are contained in a field Ω , then the compositum KL/F is in the class.

Note that (iii) follows formally from (i) and (ii).

(a)

Show that finite extensions form a distinguished class.

Proof.

Property (i) follows directly from what we have proved in class.

Assume K/F is finite.

Thus, $K = F(a_1, \dots, a_n)$ for $a_1, \dots, a_n \in K$

Thus, consider $L(K)$. This is the smallest field of L adjoined with elements of K . Thus, $KL = L(K) = L(F(a_1, \dots, a_n))$.

Since $F \subset L$ adding F does not do anything to L . Thus, $L(F(a_1, \dots, a_n)) = L(a_1, \dots, a_n)$.

Thus, KL/L is finite. ■

(b)

Show that algebraic extensions form a distinguished class.

Proof.

Property (i) follows from class.

Assume K/F is algebraic.

Consider $x \in KL$. Thus, there is a finite number of elements of K that make up x .

Call these $(a_1, \dots, a_n)$. Consider $L(a_1, \dots, a_n)/L$. Since K/F is algebraic this means that $a_1, \dots, a_n$ satisfy polynomials with coefficients in $F \subset L$. Thus, these also work for L .

Therefore, $L(a_1, \dots, a_n)/L$ is algebraic. Since $x \in L(a_1, \dots, a_n)$. This implies that x is algebraic over L . Thus, KL/L is algebraic. ■

Problem 3:

(a)

Show directly that $X^3 - 2$ is irreducible over the field $\mathbb{Q}(\omega)$, by using the arithmetic of the ring $\mathbb{Z}[\omega]$.

Proof.

Assume for contradiction that there exists a nontrivial factorization

$$X^3 - 2 = (X - \alpha)(X^2 + BX + C)$$

s.t. $\alpha, B, C \in \mathbb{Z}[\omega]$

Then, $\alpha^3 = 2$.

Thus, $N(\alpha)^3 = N(2) = 4$.

This implies that $N(\alpha) = 4^{1/3} \notin \mathbb{Z}$

Thus, this is a contradiction so, $X^3 - 2$ is irreducible in $\mathbb{Z}[\omega][X]$. Therefore, by Gauss' Lemma since $\mathbb{Z}[\omega]$ is a UFD and $\mathbb{Q}[\omega]$ is its fraction field. $X^3 - 2$ is also irreducible over $\mathbb{Q}[\omega]$. ■

(b)

Find the irreducible polynomial (over $\mathbb{Q}$) satisfied by $\alpha^2 + \omega\alpha$.

Proof.

Consider $\mathbb{Q}(\alpha^2 + \alpha\omega)$

Consider $(\alpha(\alpha + \omega))^3 = 6 + 6\omega(\alpha^2 + \alpha\omega)$

Since $\alpha^2 + \alpha\omega \in \mathbb{Q}(\alpha^2 + \alpha\omega)$ and $6 \in \mathbb{Q}(\alpha^2 + \alpha\omega)$. This implies that $\omega \in \mathbb{Q}(\alpha^2 + \alpha\omega)$.

Let $y = \alpha^2 + \alpha\omega$

Thus, $\alpha y = 2 + \omega\alpha^2 = 2 + \omega(y - \alpha\omega) = 2 + \omega y - \omega^2\alpha \implies \alpha(y + \omega^2) = 2 + \omega y \implies$

$\alpha = (2 + \omega y) / (y + \omega^2)$

Thus, $\alpha \in \mathbb{Q}(y)$

Thus, $\mathbb{Q}(y) = \mathbb{Q}(\alpha, \omega)$

$f(x) = x^6 + 6x^4 - 12x^3 + 36x^2 - 36x + 36$

$f(y) = 0$ you can check

Assume $f(x)$ is reducible over $\mathbb{Q}$. Then this means that y will be in one of those factors that has degree less than 6. This is a contradiction since we showed that $\mathbb{Q}(y)/\mathbb{Q}$ has degree 6.

■ Thus, $f(x)$ is irreducible over $\mathbb{Q}$.



Problem 4:**(a)**

Show that the polynomial $X^4 - X - 1$ is irreducible over $\mathbb{Q}$

Proof.

Assume $X^4 - X - 1$ is not irreducible in $\mathbb{Q}$. Since $X^4 - X - 1$ is primitive by Gauss's lemma it is also reducible over $\mathbb{Z}$.

Consider $X^4 - X - 1 = (X - r)(X^3 + aX^2 + bX + c)$ for $r, a, b, c \in \mathbb{Z}$.

Thus, $X^4 - X - 1 = X^4 + (a - r)X^3 + (b - ar)X^2 + (c - br)X - cr$

$$X^3 : a - r = 0 \implies a = r$$

$$X^2 : b - ar = 0 \implies b = r^2$$

$$X : c - br = -1 \implies c = r^3 - 1$$

$$cr = 1 \implies r^4 - r = 1$$

$$r(r^3 - 1) = 1$$

Thus, r divides both sides. This implies that $r = \pm 1$.

Both of these solutions do not work.

Thus, consider $X^4 - X - 1 = (X^2 + aX + B)(X^2 + cX + d) = X^4 + (a + c)X^3 + (ac + b + d)X^2 + (ad + bc)X + bd$

$$X^3 : a = -c$$

$$X^2 : b + d = a^2$$

$$X : a(d - b) = -1$$

$$bd = -1$$

This implies that b, d are units thus $b = 1, d = -1$ or $b = -1, d = 1$.

In either case we get

$$b + d = 0 = a^2 \implies a = 0.$$

This is a contradiction since this would imply that $0(d - b) = 0 = -1$

Thus, $X^4 - X - 1$ is irreducible over $\mathbb{Q}$. ■

(b)

Let α be a root of the equation $X^4 - X - 1$ in $\mathbb{C}$. Find the irreducible polynomial satisfied by α^2 over $\mathbb{Q}$.

Proof.

$$f(X) = X^4 - 2X^2 - X + 1$$

$$f(\alpha^2) = \alpha^8 - 2\alpha^4 - \alpha^2 + 1 = (\alpha^4 - 1)^2 - \alpha^2$$

Since $\alpha^4 - \alpha - 1 = 0 \implies \alpha = \alpha^4 - 1$

$$(\alpha^4 - 1)^2 - \alpha^2 = \alpha^2 - \alpha^2 = 0.$$

Since this is primitive we can check if it is irreducible over $\mathbb{Z}$.

Assume it is not irreducible over $\mathbb{Z}$, then

$$\text{If } X^4 - 2X^2 - X + 1 = (X - r)(X^3 + aX^2 + bX + c) = X^4 + (a - r)X^3 + (b - ar)X^2 + (c - br)X - cr$$

$$X^3 : a = r$$

$$X^2 : b = -2 + r^2$$

$$X : c = r^3 - 2r - 1$$

$$r^4 - 2r^2 - r = -1 \implies r(r^3 - 2r - 1) = -1 \text{ Thus } r \text{ divides } -1. \text{ Thus, } r = -1, 1.$$

$$\text{Then } (1)^4 - 2(1)^2 - 1 + 1 = -1 \neq 0 \text{ and } (-1)^4 - 2(-1)^2 + 1 + 1 = 1 \neq 0$$

$$\text{If } X^4 - 2X^2 - X + 1 = (X^2 + aX + b)(X^2 + cX + d) = X^4 + (a + c)X^3 + (ac + b + d)X^2 + (ad + bc)X + bd$$

$$X^3 : a = -c$$

$$X^2 : b + d = a^2 - 2$$

$$X : a(d - b) = -1$$

$$bd = 1$$

$$\text{Thus, } b = d = -1, 1$$

$$\text{This is a contradiction since } a(d - b) = -1 \implies 0 = -1$$

Thus, $f(X)$ is irreducible over $\mathbb{Z}$ thus also over $\mathbb{Q}$ since it is primitive. ■

Problem 5:

Let $K = \mathbb{Q}(\alpha, i)$ where α is a root of the polynomial $X^4 - 2$.

(a)

Find the degree of $K/\mathbb{Q}$.

Proof.

First this polynomial is irreducible by Eisenstein's Criterion using $p = 2$.

Thus, $\mathbb{Q}(\alpha)/\mathbb{Q}$ has degree 4.

Let $\alpha = \sqrt[4]{2}$. Then we know that $\mathbb{Q}(\alpha)$ is a real number field, and so $X^2 - 1$ which only has complex roots is also not irreducible.

Thus, $\mathbb{Q}(\alpha, i)/\mathbb{Q}(\alpha)$ has degree 2.

Therefore, $K/\mathbb{Q}$ has degree 8. ■

(b)

Find the irreducible polynomial over $\mathbb{Q}$ satisfied by $\alpha + i$.

Proof.

First consider $\mathbb{Q}(\alpha + i)$. Let $z = \alpha + i$

$$(z - i)^4 = 2 \implies z^4 - 4z^3i - 6z^2 + 4zi + 1 = 2 \implies i = (z^4 - 6z^2 - 1)/(4(z^3 - z))$$

Thus, $i \in \mathbb{Q}(z)$. Thus, $\alpha \in \mathbb{Q}(z)$. Thus, $\mathbb{Q}(z) = \mathbb{Q}(\alpha, i)$

$$f(x) = x^8 + 4x^6 + 2x^4 + 28x^2 + 1$$

$$f(\alpha + i) = 4 + 16i\alpha^3 - 48i\alpha^2 + 64i\alpha + 24 - 16i\alpha^3 + 38\alpha^2 + 64i\alpha - 28 = 0$$

Assume that $f(x)$ is reducible over $\mathbb{Q}$. This is a contradiction since this would imply that $\alpha + i$ would be in a proper factor that has degree less than 8. However $\mathbb{Q}(\alpha + i)/\mathbb{Q}$ has degree 8.

Thus, $f(x)$ is irreducible. ■